Guillermo Fernando Farfán Romero

Senior Python Backend Engineer & Technical Lead

Buenos Aires / Salta, Argentina (UTC-3 · Open to Remote / Hybrid / Relocation) | +54 9 11 2187-1473 | fernando.farfan16@gmail.com
gfarfan.dev | linkedin.com/in/guillermo-farfan | github.com/Fernandofarfan | credly.com/users/fernando-farfan.855a5067

PROFESSIONAL SUMMARY

Senior Python Backend Engineer and Technical Lead with 8+ years of experience designing scalable microservices architectures, low-latency RESTful APIs, financial transaction engines, and enterprise integration backends. Specialist in **Python 3.10+ (FastAPI, Asyncio, uvloop, Celery, SQLAlchemy 2.0, Pydantic v2, Django REST Framework)**, complemented by **Java (Spring Boot)** and **TypeScript / Node.js**, optimized persistence in **PostgreSQL (with pgvector), SQL Server, and Redis**, and cloud-native deployments on **Google Cloud Platform (GKE Autopilot, Cloud Run)**.

Proven track record driving the adoption of **Clean Architecture and Domain-Driven Design (DDD)**, automated test suites with **Pytest (>85% coverage)** under TDD, distributed observability (**IBM Instana APM, OpenTelemetry**), and cloud financial optimization (**FinOps**).

TECHNICAL SKILLS & COMPETENCIES

- **Languages & Backend Core:** Python 3.10+ (Asyncio, Strict Type Hints, Concurrency, uvloop, Finite State Machines - FSM), Java (Spring Boot, JPA/Hibernate), TypeScript / Node.js, PowerShell & Bash Scripting.
- **Python Frameworks & Libraries:** FastAPI, Pydantic v2, SQLAlchemy 2.0 (Async Engine & ORM), Django / Django REST Framework (DRF), Celery, Alembic (Zero-Downtime Migrations), Flask, httpx, Uvicorn, Gunicorn, Pandas, Polars, PyArrow, GenAI / LLM Integration (Vertex AI, Gemini / OpenAI API, RAG, pgvector), Streaming APIs (SSE / WebSockets).
- **Databases, Caching & Messaging:** PostgreSQL (Declarative Partitioning, pgvector, JSONB, asyncpg, PgBouncer), Redis (Distributed Caching, Rate Limiting, Pub/Sub, Celery Broker), Microsoft SQL Server (2012–2022, T-SQL), Google Cloud Pub/Sub, BigQuery, MySQL, MongoDB.
- **Cloud, Containers & DevOps:** Google Cloud Platform (Cloud Run, GKE Autopilot, Compute Engine, M4CE, Cloud Storage, Secret Manager, Cloud Armor, Cloud KMS), Docker (Multi-stage builds), Docker Compose, Terraform, ArgoCD, GitHub Actions, GitLab CI.
- **Architecture, Quality & Observability:** Microservices, Clean Architecture, Hexagonal Architecture (Ports & Adapters), Domain-Driven Design (DDD), Event-Driven Architecture (EDA, Dead Letter Queues - DLQ), Architecture Decision Records (ADRs), OpenAPI/Swagger, TDD/BDD (Pytest, Factory Boy, Jest), OpenTelemetry (OTel), IBM Instana APM, Zabbix, SonarQube, Trivy.

PROFESSIONAL EXPERIENCE

Technical Lead & Senior Systems Engineer (Python Backend Focus) | Reforest Latam

May 2026 – Present | Full-time (Remote)

- Led the adoption of **Clean Architecture and Domain-Driven Design (DDD)** across an **engineering team of 6** for environmental telemetry, ERP, and financial automation platforms (featuring **FSM-based transactional bots and n8n reconciliation**), formalizing ADRs and accelerating **modular feature delivery by 35%**.
- Orchestrated asynchronous microservices and distributed workers using **FastAPI, SQLAlchemy 2.0, Pydantic v2, Celery, and Cloud Pub/Sub**, paired with a **Redis** distributed caching layer (**-45% concurrent query latency**) and **100% zero-downtime Alembic migrations** on PostgreSQL.
- Established **>85% automated test coverage with Pytest and Jest** under TDD practices, and implemented **GitOps pipelines in GitHub Actions and ArgoCD** targeting **Cloud Run and GKE Autopilot** (**Lead Time cut from 5 days to minutes**, **100% of production images free of critical/high CVEs via Trivy/SonarQube**).
- Spearheaded **FinOps and Zero-Trust security on GCP** (dedicated Service Accounts, least-privilege IAM, Cloud Armor WAF, JWT/OAuth2), achieving a **sustained 28% reduction in monthly cloud spend**.

Senior Python, Cloud & Software Consultant | Independent Consulting & Enterprise Projects

January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)

- **Ecovalores / Matriz OMS, SeSocio & Wenance / Falabella (Fintech & Trading):** Engineered high-concurrency **FastAPI** middleware for **Matriz OMS** trading protocols, credit-scoring microservices, and low-latency REST APIs in **Django REST Framework**; eliminated N+1 query bottlenecks using *SQLAlchemy 2.0 (*selective eager loading*), Pydantic v2, and Redis**, achieving a** >60% reduction in p95 latency on critical endpoints*.
- **Ganfeng Lithium (Industrial Automation on GCP):** Developed ingestion microservices and data pipelines in **Java (Spring Boot) and Python** to synchronize logistics telemetry metrics and remote operational reporting on Google Cloud Platform.
- **Grupo Ramos (75-Server Cloud Migration & SAP BW on HANA Azure → GCP):** Automated GCP provisioning with **Terraform and Python** (Shared VPC, GKE, Cloud KMS) for the migration of **68 corporate VMs (Phases I & II via M4CE: NCF Electronic Invoicing, Self-Checkout, Planograms, Azul SFTP 5.4 TB)** and **7 SAP BW on HANA servers from Azure to GCP** (m2-ultramem-208, **Hyperdisk Extreme**, **~2.9 TB ingestion to GCS via SAP Backint / Parallel Composite Uploads**, and **PSC to BigQuery**).
- **Junta de Beneficencia de Guayaquil & Universal Group (APM & Automation):** Deployed full-stack observability with **IBM Instana APM, OpenTelemetry, and Zabbix (-45% MTTR)** and authored **PowerShell/Bash (dracut/nmcli)** automation scripts for migrating **20 RHEL 8 / Windows pilot VMs** to Universal Group's **Hyper-V 2025 cluster (55 TB Alletra SAN)**.

Senior Software Engineer & Database Administrator (DBA) | University of Buenos Aires (UBA)

March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)

- Engineered concurrent asynchronous **Python, SQLAlchemy, and Bash** microservices and ETL synchronization pipelines across **8+ TB of institutional data (+500 GB migrated from critical legacy databases)** serving **+50,000 users (+10,000 concurrent active users)**, reducing **manual operational effort by 60%** and batch windows from hours to minutes.
- Implemented schema version control with **Alembic** and tuned complex queries on **PostgreSQL and SQL Server**, achieving a **55% reduction in p95 query latency (-40% average response time)** with **RPO < 15 min and RTO < 1 hour**.

Systems Infrastructure & Operations Coordinator | Government of the City of Buenos Aires (GCBA)

January 2020 – December 2023 | Ministry of Economic Development (Full-time / Buenos Aires, Argentina)

- Established technical interoperability standards and automated backend validation layers that reduced **cross-departmental data inconsistencies by 40%**, maintaining a **99.9% uptime SLA** and **-30% production incidents** for **100,000+ monthly users** across **SQL Server, PostgreSQL, MySQL, and MongoDB**.

FEATURED PROJECTS & HACKATHONS

- **Autonomous FinOps Engine (Aleph Hackathon 2026):** Autonomous **Python** engine that scans Google Cloud environments to detect idle resources and automatically generates **Terraform (IaC)** remediation code.
- **OmniSync (Event-Driven Architecture - EDA) & Monitor CI/CD (DORA):** Polyglot engine featuring eventual consistency, compensating events, and **Dead Letter Queues (DLQ)**, plus a real-time TypeScript/Node.js/WebSockets CI/CD monitoring platform.

EDUCATION, CERTIFICATIONS & LANGUAGES

- **Education: Associate Degree in Information Systems** | UTN *(In Progress — Expected Graduation: Dec 2027)* · **Bachelor's Studies in Labor Relations & HR** | UBA *(Advanced University Coursework Completed / Organizational Leadership)*.

  - *Certifications* (Verifiable on Credly & Coursera):* Meta Back-End Developer Certificate | IBM DevOps and Software Engineering Professional Certificate | IBM Java Developer Professional Certificate (V2) | IBM JavaScript Backend Developer (V2) | IBM Full Stack Software Developer (V5) | Meta Full-Stack Engineer | Google Data Analytics.
- **Languages: Spanish:** Native · **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, and documentation).